

1. Data Cleaning & Modeling:

- Fix missing fuel consumption values (use mean imputation).

Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status
1	null	null	null	null	null	null	null
2	null	null	null	null	null	null	null
3	null	null	null	null	null	null	null
4	null	null	null	null	null	null	null
5	null	null	null	null	null	null	null
6	null	null	null	null	null	null	null
7	null	null	null	null	null	null	null
8	null	null	null	null	null	null	null
9	null	null	null	null	null	null	null
10	null	null	null	null	null	null	null
11	null	null	null	null	null	null	null
12	null	null	null	null	null	null	null
13	null	null	null	null	null	null	null
14	null	null	null	null	null	null	null
15	null	null	null	null	null	null	null
16	null	null	null	null	null	null	null
17	null	null	null	null	null	null	null
18	null	null	null	null	null	null	null
19	null	null	null	null	null	null	null
20	null	null	null	null	null	null	null

Vehicle_ID	Vehicle_Type	Capacity_kg	Maintenance_Cost
V01	Van	8424	5633
V02	Truck	1970	6776
V03	Truck	1207	18031
V04	Mini-Truck	8803	9033
V05	Truck	9941	17751
V06	Van	6053	5914
V07	Van	2598	12837
8	null	null	null
9	null	null	null
10	null	null	null
11	null	null	null
12	null	null	null
13	null	null	null
14	null	null	null
15	null	null	null
16	null	null	null
17	null	null	null
18	null	null	null
19	null	null	null
20	null	null	null
21	null	null	null
22	null	null	null
23	null	null	null

Replace Values

Replace one value with another in the selected columns.

Value To Find
null

Replace With
91.88

OK Cancel

Time	Status
85.04	Late
58.08	On-Time
5.24	On-Time
16.22	On-Time
105.21	Late
31.17	On-Time
51.77	On-Time
188.52	Late
104.51	On-Time
179.88	On-Time
null	Late
102.91	On-Time
107.23	On-Time
155.52	On-Time
148.87	On-Time

2. DAX Measures:

- Fuel Efficiency = Distance / Fuel Consumed

1 Fuel Efficiency = $\text{DIVIDE}(\text{SUM}(\text{Trip_Data}[\text{Distance_km}]), \text{SUM}(\text{Trip_Data}[\text{Fuel_Consumed_L}]), 0)$

Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date
V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023
V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023
V06	D08	Mumbai	Pune	1459	154.7	On-Time	17 February 2023
V04	D09	Hyderabad	Pune	382	26.6	On-Time	18 February 2023
V06	D08	Pune	Mumbai	398	33.2	On-Time	15 February 2023
V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023
V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023
V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023
V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023
V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023
V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023
V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023
V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023

On-Time Delivery % = On-Time Trips / Total Trips:

1 Total Trips = $\text{COUNTS}(\text{Trip_Data})$

Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date
V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023
V06	D08	Mumbai	Pune	1459	154.7	On-Time	17 February 2023
V04	D09	Hyderabad	Pune	382	26.6	On-Time	18 February 2023
V06	D08	Pune	Mumbai	398	33.2	On-Time	15 February 2023
V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023
V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023
V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023
V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023
V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023
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V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023
V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023
V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023
V05	D06	Bangalore	Mumbai	858	91.88	Late	18 January 2023

1 On Time Trips = $\text{CALCULATE}(\text{COUNTS}(\text{Trip_Data}[\text{Delivery_Status}]))$

Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date
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V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023

1 On-Time Deliver% = $\text{DIVIDE}(\text{Trip_Data}[\text{Total Trips}], \text{Trip_Data}[\text{On Time Trips}], 0)$

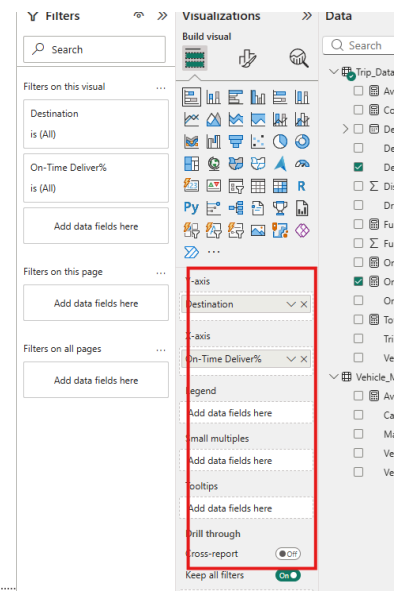
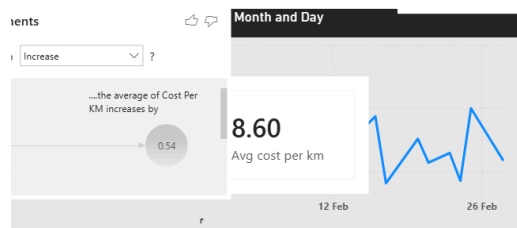
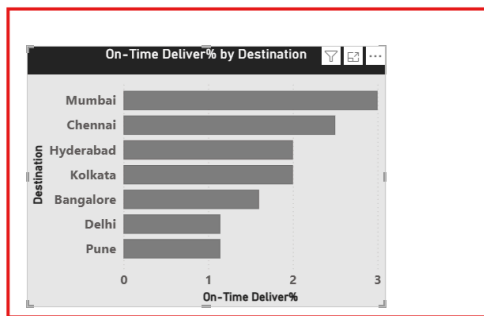
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Cost per km = (fuel cost + Maintenance Cost) / Distance Note fuel cost =100

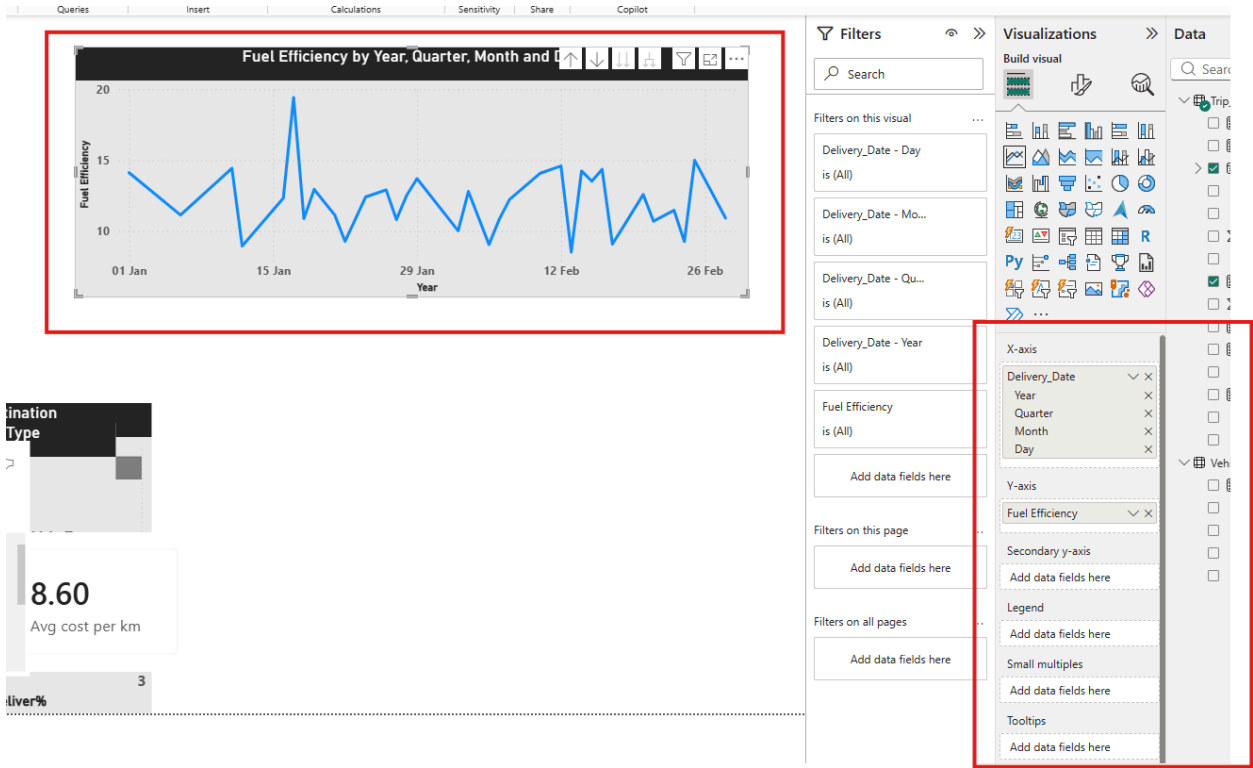
Formula Bar							
1 Cost Per KM = DIVIDE(100*SUM(Trip_Data[Fuel_Consumed_L])+SUM(Vehicle_Master[Maintenance_Cost]),SUM(Trip_Data[Distance_km]),0)							
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3. Visualization:

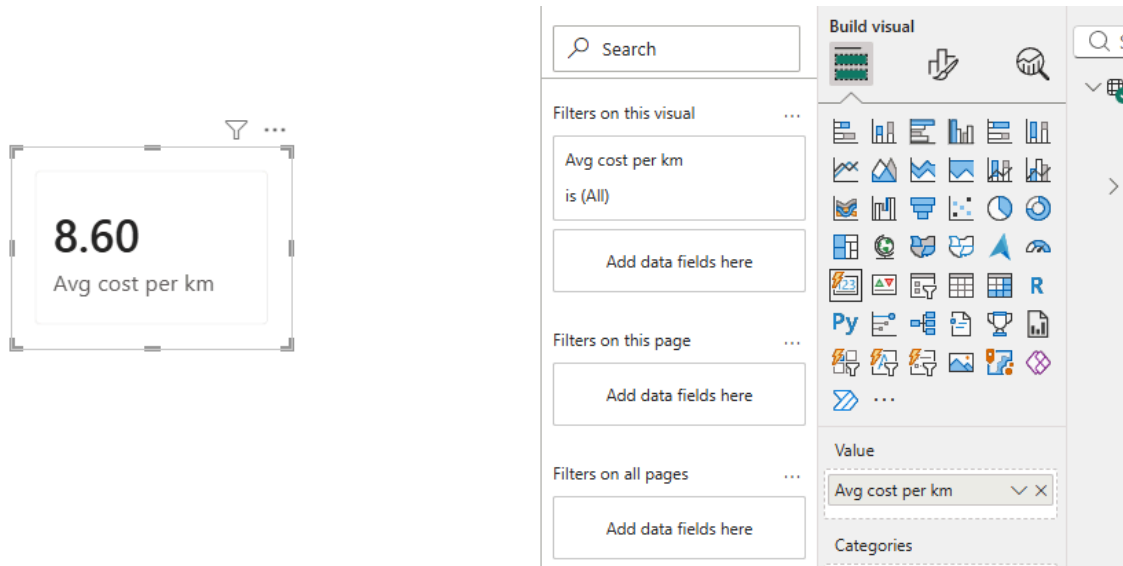
Bar chart: On-Time Delivery % by Destination.



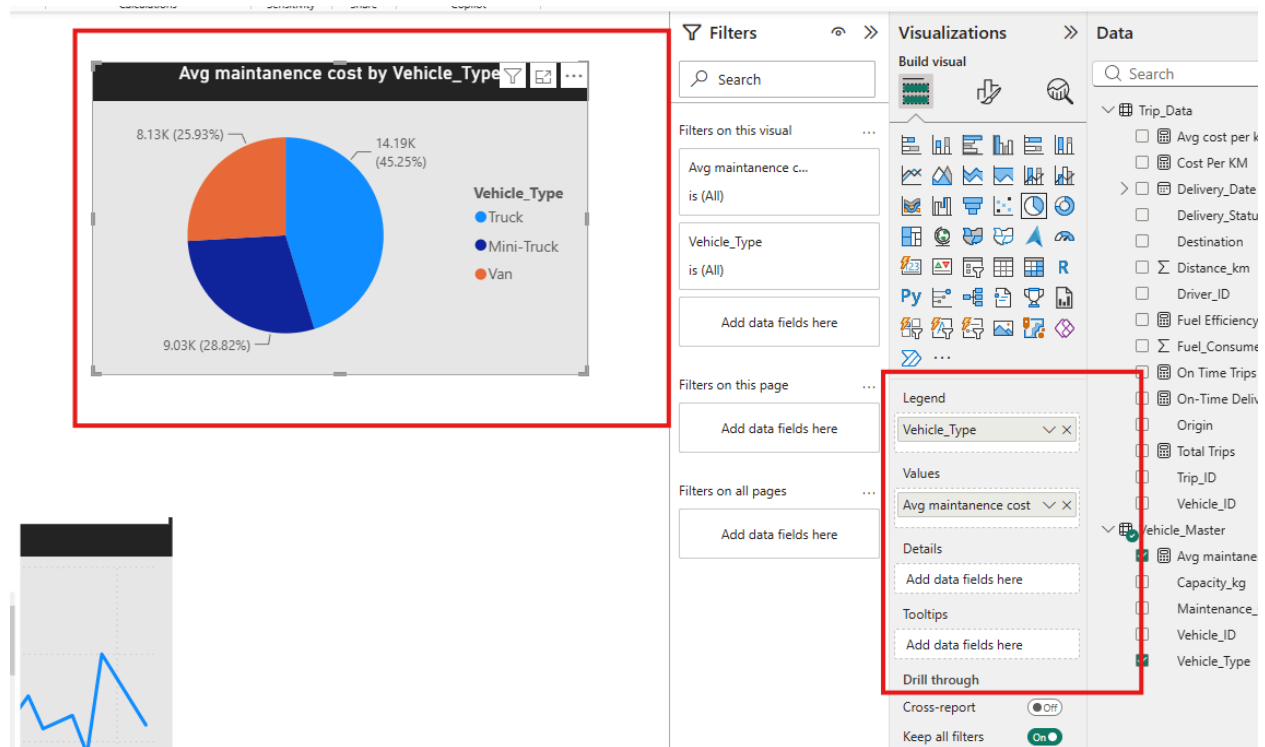
Line chart: Fuel efficiency trend by delivery date.



Cards visualization:
Average cost per km

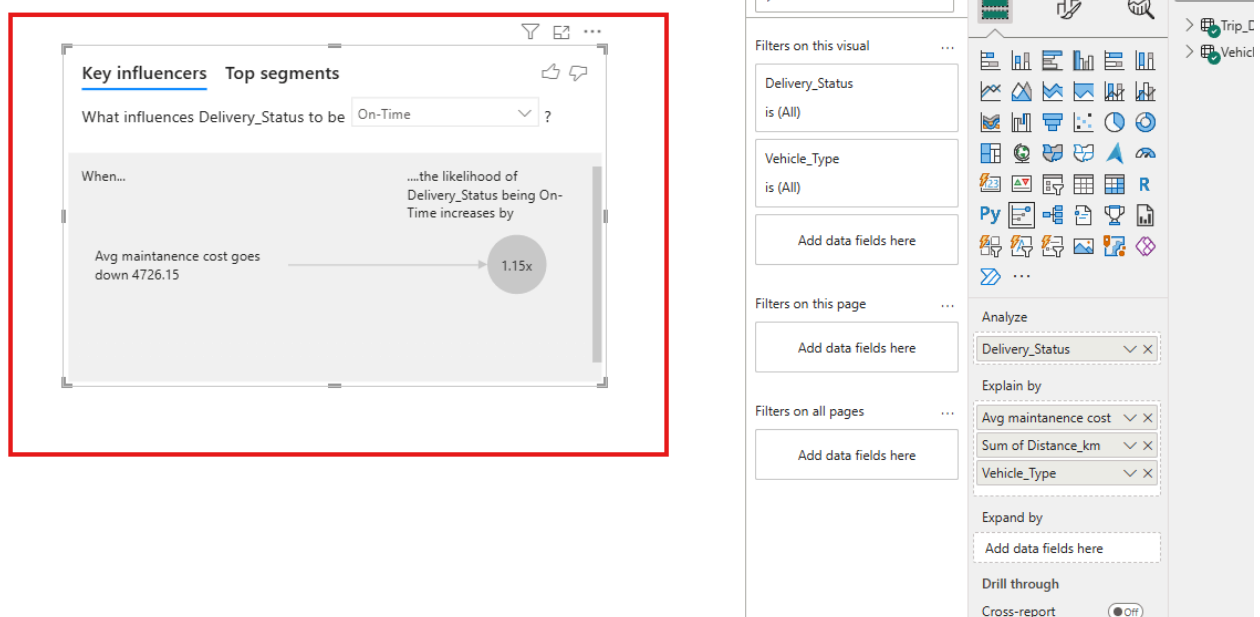


Pie chart: vehicle type vs Average maintenance cost



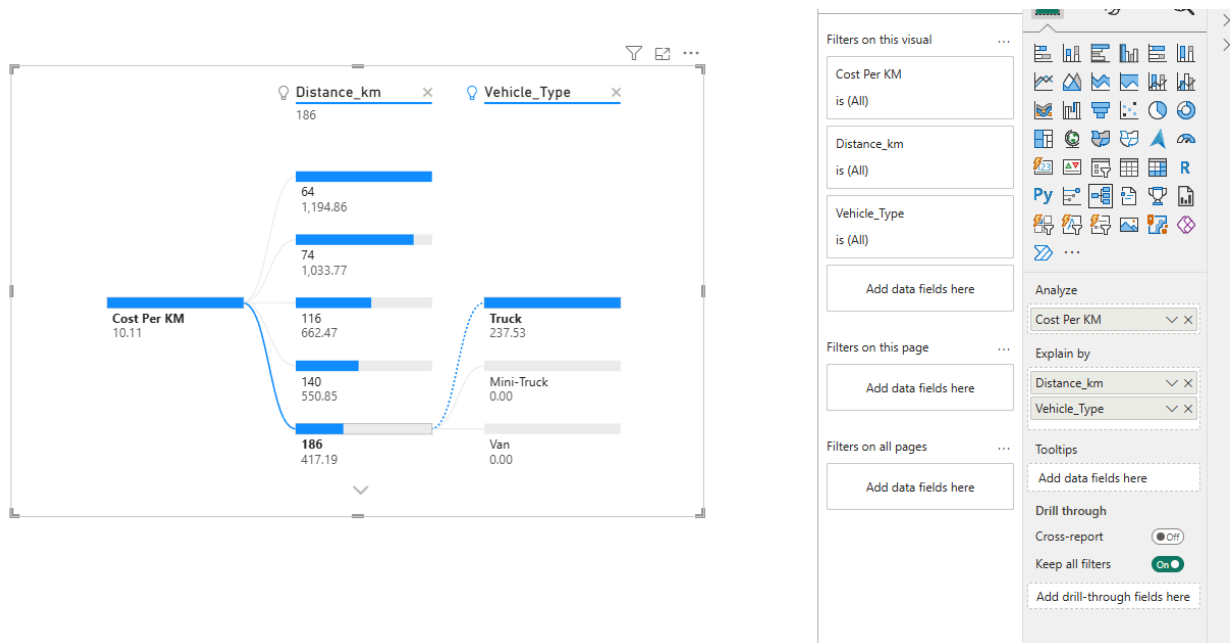
4. AI-Powered Visuals

Key Influencers Visual: ➤ Add Delivery Status in the Analyze and explain by -distance in km, vehicle_type, Driver_ID

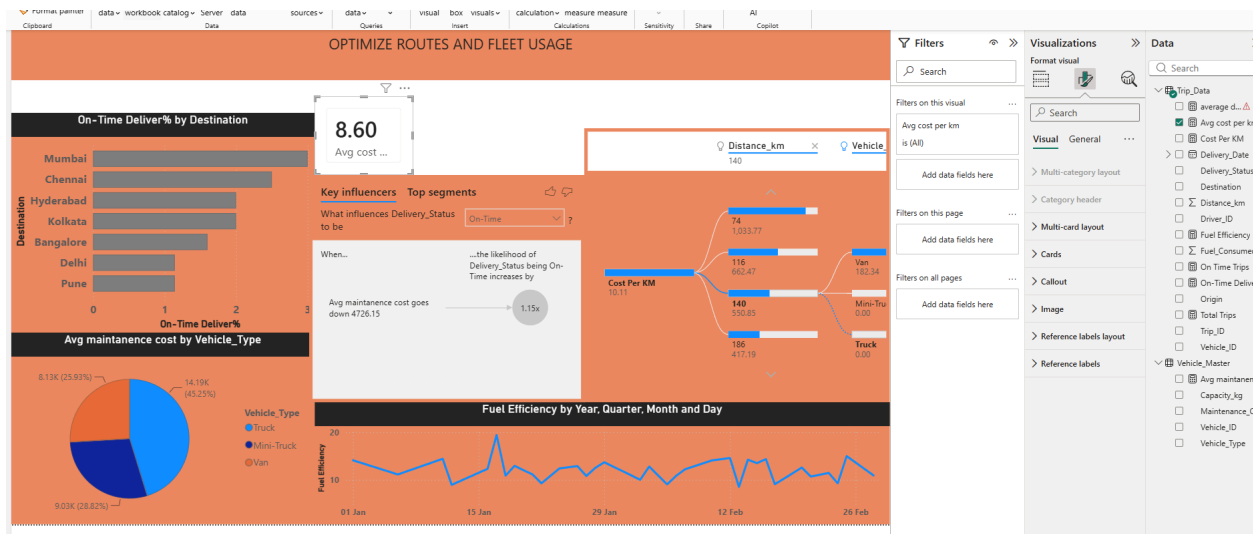


Decomposition Tree (AI Visual):

➤ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.



DASHBOARD



Observations:

On-Time Delivery: Shows how many deliveries were on time.

Fuel Efficiency: Shows how efficiently vehicles use fuel

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Cost per Km: Shows the cost of travelling one kilometer.

Vehicle Type: Shows which vehicle type has higher maintenance cost

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Delivery Trend: Shows how fuel efficiency changes by date.

Driver Performance: Shows factors that affect delivery status.

Cost Analysis: Shows which vehicle and distance have higher costs

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Overall: The dashboard helps to **reduce cost, save fuel, and improve delivery performance.**